

PARAKOU, Benin

WMO: 653300

Lat: 9.350N

Lon: 2.617E

Elev: 393

StdP: 96.69

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.4	20.3	-5.5	2.5	25.4	-3.4	3.0	26.0	6.3	23.5	5.5	24.3	2.0	40	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.3	37.1	22.0	36.2	21.9	35.5	21.7	26.0	32.2	25.6	31.6	25.3	31.0	2.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.3	29.0	24.1	19.9	28.6	23.7	19.5	28.2	83.1	32.5	81.3	31.9	80.0	31.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.2	4.6	DB	17.1	39.3	1.3	1.0	16.2	40.1	15.4	40.7	14.7	41.3	13.8	42.0	
			WB	8.3	27.3	0.9	1.2	7.6	28.1	7.1	28.8	6.5	29.5	5.9	30.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	27.0	29.1	30.2	29.3	27.7	26.4	25.4	25.0	25.4	26.4	27.6	27.0
	DBStd	2.06	1.82	1.50	1.24	1.57	1.43	1.30	1.14	1.06	1.05	1.08	1.00	1.30
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6276	527	536	627	578	548	492	477	464	463	509	527	527
	CDD18.3	3234	269	303	369	328	290	242	218	206	213	251	277	269
	CDH23.3	34504	3172	4128	5148	4182	3276	2223	1473	1173	1486	2189	3032	3023
	CDH26.7	15446	1558	2281	2921	2147	1395	711	314	186	346	743	1406	1439
(14) Wind	WSAvg	2.0	2.1	2.1	2.3	2.5	2.3	2.0	1.9	1.9	1.6	1.6	1.8	2.2
(15) Precipitation	PrecAvg	1179	2	9	35	82	133	170	192	214	216	97	10	2
	PrecMax	1615	32	68	136	219	335	327	351	447	360	215	113	44
	PrecMin	642	0	0	0	13	48	38	57	51	70	18	0	0
	PrecStd	223	6	17	32	44	59	67	80	83	61	51	23	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	37.8	38.5	37.8	35.2	33.0	31.2	30.6	31.4	33.2	35.2	35.4
		MCWB	18.6	20.9	21.9	23.3	24.9	25.0	24.6	24.6	24.8	23.6	20.9	18.8
	2%	DB	34.9	36.5	37.3	36.5	33.8	31.8	30.1	29.4	30.4	32.1	34.2	34.3
		MCWB	18.6	20.8	22.0	23.5	25.0	24.8	24.3	24.3	24.6	24.0	20.7	19.1
	5%	DB	33.9	35.6	36.4	35.4	32.8	31.0	29.3	28.6	29.5	31.2	33.4	33.4
		MCWB	18.9	20.5	22.2	24.1	25.0	24.6	24.1	24.1	24.4	24.2	21.0	19.2
	10%	DB	32.9	34.8	35.7	34.2	32.0	30.1	28.5	27.7	28.7	30.3	32.5	32.5
		MCWB	18.5	20.3	22.3	24.4	24.9	24.5	23.9	23.8	24.2	24.3	21.3	19.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	25.2	25.8	26.5	26.6	26.0	25.2	25.2	25.7	26.0	25.6	24.3
		MCDB	31.8	33.8	33.9	33.2	32.2	30.9	29.5	29.2	29.9	30.6	30.7	30.8
	2%	WB	23.1	24.4	25.2	25.9	26.0	25.4	24.8	24.7	25.1	25.4	24.9	23.5
		MCDB	30.2	32.1	33.2	33.2	31.9	30.3	28.8	28.4	29.2	30.1	30.7	29.9
	5%	WB	22.3	23.7	24.7	25.5	25.6	25.0	24.5	24.2	24.7	25.1	24.3	22.8
		MCDB	29.3	31.1	32.5	32.6	31.3	29.9	28.4	27.7	28.6	29.7	29.8	29.1
	10%	WB	21.3	23.1	24.3	25.1	25.2	24.7	24.1	23.9	24.4	24.7	23.7	22.0
		MCDB	28.8	30.7	31.7	32.0	30.8	29.3	27.8	27.2	28.1	29.2	28.9	28.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.8	11.3	10.0	8.4	7.2	6.1	5.6	6.7	8.2	11.1	12.4
		MCDBR	13.3	12.7	12.6	11.7	9.7	8.5	7.4	6.9	8.1	9.6	12.1	13.1
		MCWBR	6.5	5.6	4.0	3.1	3.1	2.9	2.7	2.7	3.4	3.6	4.1	5.7
	5% WB	MCDBR	11.2	11.4	11.0	10.0	8.6	7.7	6.8	6.3	7.5	8.5	9.8	11.1
		MCWBR	6.4	4.8	3.4	3.3	3.4	3.0	2.8	2.7	3.5	3.7	3.5	5.2
(40) Clear-Sky Solar Irradiance	taub		0.709	0.757	0.803	0.706	0.607	0.554	0.508	0.495	0.507	0.588	0.593	0.613
	taud		1.603	1.538	1.475	1.680	1.855	1.996	2.167	2.206	2.152	1.893	1.821	1.774
	Ebn at Noon		642	631	612	668	721	751	791	814	814	743	725	702
	Edh at Noon		265	291	313	252	207	177	150	147	157	201	211	219
(44) All-Sky Solar Radiation	RadAvg		5.23	5.43	5.76	5.81	5.54	5.03	4.37	4.04	4.61	5.20	5.53	5.21
	RadStd		0.15	0.16	0.23	0.16	0.22	0.20	0.19	0.25	0.23	0.24	0.16	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page